

Document Intelligence Platform

- End-to-end extraction, validation and reconciliation for financial documents.
- Live application: document-intelligence-platform-kace.onrender.com
- FastAPI | Gemini vision | Evidence-backed JSON

Objective and scope

- Four document types: invoice, balance sheet, profit and loss, cash flow statement.
- PDF/JPG/PNG uploads, maximum three pages and 15 MB, validated before OCR or AI.
- Meaningful fields and tables with source snippets and page evidence.
- Document-type financial formulas with PASS, FAIL and NOT_APPLICABLE.

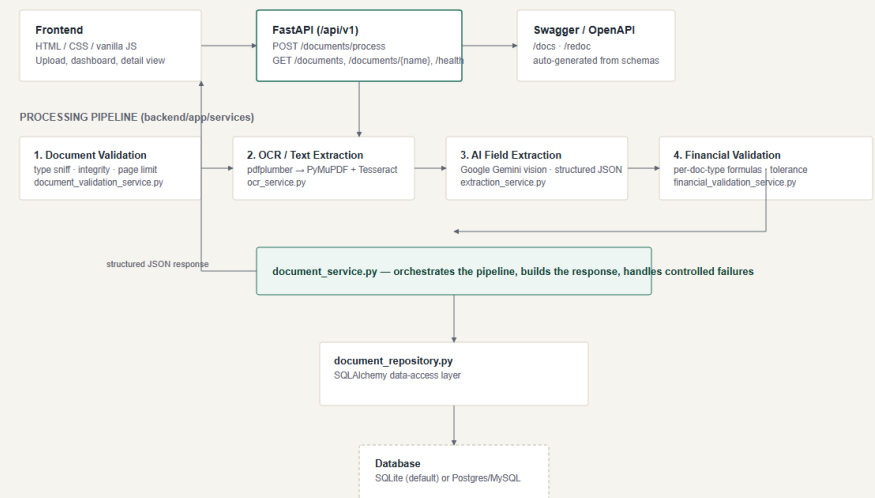
System architecture

Browser dashboard -> FastAPI -> validation ->
OCR -> Gemini vision -> financial validation ->
repository -> database.

See architecture.png and architecture.pdf for
the complete flow and service boundaries.

Document Intelligence Platform — Architecture

Upload → validate → OCR/extract → AI field extraction → financial validation → persist → dashboard/API



Validation, OCR and input safety

- File signatures, type, integrity, readability, page count and size are checked from bytes.
- Native PDF text is preferred; scanned pages use PyMuPDF rasterization and Tesseract fallback.
- Page-tagged text and normalized images preserve provenance for every field.

Evidence-backed AI extraction

- google-genai SDK, strict JSON envelope, line items, evidence and confidence.
- Missing or unreadable values stay null; no unsupported inference.
- Response variants, empty output, JSON fences, retry and model fallback are handled.

Financial validation engine

- Invoice: $\text{subtotal} + \text{tax} - \text{discount} = \text{total}$.
- Balance sheet: $\text{liabilities} + \text{equity} = \text{assets}$.
- Profit and loss: $\text{revenue} - \text{cost of sales} = \text{gross profit}$.
- Cash flow: $\text{operating} + \text{investing} + \text{financing} + \text{FX} = \text{net change}$.
- Rounding tolerance is explicit; missing operands produce NOT_APPLICABLE.

Product experience and API

- Responsive dashboard, upload workflow, search, filters, statistics and refresh.
- Centered document workspace with Fields, Validation and Raw JSON tabs.
- POST /api/v1/documents/process; GET /api/v1/documents; GET /api/v1/documents/{name}; GET /api/v1/health; /docs.

Verification and deployment

- 25 automated tests pass without a network call or API key.
- Fresh live invoice verification returned file validation PASS and processing PASS.
- The live test returned 43 extracted field entries and 6 line items.
- Public Render deployment, health endpoint, Swagger and four-type sample fixtures are included.

Limitations and production plan

- Render Free local SQLite storage is ephemeral; managed Postgres is required for durable history.
- Production adds authentication, tenant isolation, object storage, durable workers, observability and evaluations.

AI usage and submission deliverables

- AI coding assistance was reviewed and test-verified; Gemini is used only for configured, evidence-backed extraction.
- Repository includes source, live links, README, tests, samples, architecture source/PNG/PDF and this presentation.